Answer Key

Quiz 1 - Retake

Quiz Code: **051036**

Quiz Information

Total Questions:	**29**
Time Limit:	**15 minutes**
Quiz Period:	**Mar 12, 2026 13:40 - Mar 12, 2026 14:40**
Answer Key Released:	**March 19, 2026 12:59**

All Questions with Correct Answers

Question 1

1. In the context of the loss surface $J(\theta)$, what does a "global minimum" represent?

A **The absolute lowest loss across the parameter space**

Correct Answer

B The point where the gradient is steepest

C A local valley that is not the lowest point

D The starting point of the algorithm

Question 2

2. Which direction does the gradient vector $\nabla J(\theta)$ always point?

A Toward the global minimum

B **Steepest ascent (uphill)** Correct Answer

C Steepest descent (downhill)

D Orthogonal to the loss surface

Question 3

3. In Gradient Descent, what happens if the learning rate α is too large?

A Convergence takes too long

B It always finds the global minimum

C **It may overshoot and fail to converge** Correct Answer

D The gradient becomes zero

Question 4

4. What is a partial derivative $\frac{\partial J}{\partial \theta_i}$ conceptually?

A Total change in J when all parameters change

B **Slope of the loss with respect to one parameter** Correct Answer

C The average of all gradients

D The second-order derivative

Question 5

5. When implementing KNN in NumPy using broadcasting for subtraction $(X - Y)$, what shape manipulation is typically required?

A **Reshaping X to $(M, 1, D)$ and Y to $(1, N, D)$**

Correct Answer

B Flattening both into 1D arrays

C Transposing Y so both are (N, D)

D No reshaping is needed

Question 6

6. Why is broadcasting preferred over for-loops in KNN?

A Uses less memory

B **Leverages vectorized operations for speed**

Correct Answer

C Handles missing data automatically

D Reduces complexity from $O(N)$ to $O(1)$

Question 7

7. In Polynomial Regression, increasing the degree of the polynomial generally:

A Keeps complexity the same

B Decreases complexity

C **Increases complexity/flexibility** Correct Answer

D Converts it to logistic regression

Question 8

8. Linear Regression is "Linear" because it is linear with respect to:

A Input features x

B Target variable y

C **Model parameters θ** Correct Answer

D Error terms ϵ

Question 9

9. In Bias-Variance Decomposition, "Bias" refers to:

- A **Error from over-simplifying a complex problem** ✓ Correct Answer
- B Sensitivity to training set fluctuations
- C Noise inherent in the data
- D Difference between train and test error

Question 10

10. A model that "overfits" the training data typically has:

- A High Bias and Low Variance
- B **Low Bias and High Variance** ✓ Correct Answer
- C High Bias and High Variance
- D Low Bias and Low Variance

Question 11

11. As model complexity increases, what happens to Bias and Variance?

A Bias increases, Variance decreases

B **Bias decreases, Variance increases** Correct Answer

C Both increase

D Both decrease

Question 12

12. Geometrically, why does Lasso (L1) produce sparse solutions?

A Constraint region is a circle

B **Constraint region is a diamond with corners on the axes** Correct Answer

C It penalizes the square of weights

D It increases the learning rate

Question 13

13. What is the geometry of the L2 (Ridge) constraint region?

A **A hypersphere (circle in 2D)** Correct Answer

B A square/diamond

C A flat plane

D An infinite line

Question 14

14. What is the primary mathematical difference between L1 and L2 penalties?

A **L1 uses absolute values; L2 uses squares** Correct Answer

B L1 is for classification; L2 is for regression

C L1 uses squares; L2 uses absolute values

D L2 is always larger than L1

Question 15

15. In Support Vector Machines, what is the "Margin"?

A **Distance between boundary and nearest points**

Correct Answer

B Total number of misclassified points

C Area where loss is non-zero

D The learning rate

Question 16

16. What is the goal of a "Hard Margin" SVM?

A Allow misclassifications for generalization

B **Perfectly separate data with no violations** Correct Answer

C Minimize L1 norm of weights

D Project data into lower dimensions

Question 17

17. Which of these functions is non-linear?

A $f(x) = ax + b$

B $f(x) = w_1x_1 + w_2x_2$

C $f(x) = \sin(x)$ Correct Answer

D $f(x) = \sum \theta_i x_i$

Question 18

18. If a loss surface is "convex," what is true?

A It has many local minima

B **Any local minimum is the global minimum** Correct Answer

C The gradient is constant

D It is impossible to optimize

Question 19

19. The "Gradient Vector" is composed of:

A The sum of all features

B **All partial derivatives of the function** Correct Answer

C The weights after convergence

D Ratio of train error to test error

Question 20

20. In the Bias-Variance tradeoff, "Irreducible Error" is:

A Error from algorithm choice

B **Noise that cannot be removed by any model** Correct Answer

C Error from insufficient data

D Difference between L1 and L2

Question 21

21. For a weight vector w with n elements, the L1 norm $||w||$ is defined as:

A $\sqrt{\sum_{i=1}^{n} w_i^2}$

B $\sum_{i=1}^{n} |w_i|$ Correct Answer

C $(\sum_{i=1}^{n} w_i)^2$

D $\max(|w_i|)$

Question 22

22. Why is Polynomial Regression still a "Linear Model"?

A Graph is a straight line

B **It is linear in the parameters θ** Correct Answer

C Uses ReLU activation

D It cannot fit curves

Question 23

23. In the context of the Margin, what are "Support Vectors"?

A Average of all points

B **Data points closest to the decision boundary** Correct Answer

C Weights with the highest value

D Points far from the boundary

Question 24

24. A "Soft Margin" SVM is used when:

A Data is perfectly separable

B **There is noise or overlapping classes** Correct Answer

C Using L1 instead of L2

D Dataset has more features than samples

Question 25

25. What is the gradient of $J(\theta) = \frac{1}{2}\theta^2$?

A $1/2$

B θ Correct Answer

C 2θ

D 1

Question 26

26. In NumPy, if A is $(5, 3)$ and B is $(3,)$, what is the shape of $A + B$?

A $(5, 3)$ Correct Answer

B $(3, 5)$

C $(8, 3)$

D Error

Question 27

27. Which regularization is best for "Feature Selection"?

A Ridge (L2)

B **Lasso (L1)** Correct Answer

C OLS

D Polynomial expansion

Question 28

28. If your model has High Variance, you should:

A Use a more complex model

B **Add more data or regularize** Correct Answer

C Decrease regularization

D Use fewer samples

Question 29

29. The loss surface of Linear Regression with MSE is:

A Multi-modal

B **Convex (bowl-shaped)** Correct Answer

C A flat plane

D Non-continuous